

The Minkowski-Sum Splitting Question Is Already Answered Negatively

Automated mathematics run `fa_banach_001`
Lane 12, agent `agent_lane_12`, model GPT5.6

11 August 2026

Status: literature already answered. The exact question in arXiv:1401.2257 has a published negative answer, with compact convex sets in $\mathbb{R}^3$, in Balashov–Repovš (arXiv:0807.3104). No novelty is claimed.

The source question

Problem 5 of Repovš–Semenov, on printed and PDF page 32 of arXiv:1401.2257, asks: if A and B are closed convex subsets of a Banach space and $C = A + B$, must there be continuous maps

$$a : C \longrightarrow A, \quad b : C \longrightarrow B, \quad c = a(c) + b(c) \quad (c \in C)?$$

Exact prior answer

Balashov–Repovš, *On the splitting problem for selections*, J. Math. Anal. Appl. **355** (2009), 277–287, arXiv:0807.3104, quotes this question on PDF page 1 and immediately states that the answer is negative in every dimension at least three.

Their Example 1.1, on PDF pages 1–2, gives compact convex sets $A, B \subset \mathbb{R}^3$. Here B is the vertical segment joining $(0, 0, 0)$ and $(0, 0, 1)$, while A is assembled from the convex hulls of the upper and lower semicircles in the planes $z = 1$ and $z = 0$, together with the opposite endpoints in the other plane. Put $C = A + B$ and $c_0 = (1, 0, 1)$.

The boundary of C contains two arcs Γ_1, Γ_2 ending at c_0 . For every c on either punctured arc the decomposition $c = a(c) + b(c)$ with $a(c) \in A$ and $b(c) \in B$ is unique. Along the first arc,

$$\lim_{c \rightarrow c_0, c \in \Gamma_1} a(c) = (1, 0, 0),$$

whereas along the second,

$$\lim_{c \rightarrow c_0, c \in \Gamma_2} a(c) = (1, 0, 1).$$

Thus no admissible a can be continuous at c_0 , and consequently no pair of continuous splitting maps can exist.

Disposition

This is the same universal question, with a stronger counterexample than is needed: the ambient Banach space is finite-dimensional and both summands are compact convex sets. The run should therefore record the target as `literature_already_answered`, not as an original counterexample.

References

- [1] D. Repovš and P. V. Semenov, *Continuous selections of multivalued mappings*, arXiv:1401.2257.
- [2] M. V. Balashov and D. Repovš, *On the splitting problem for selections*, J. Math. Anal. Appl. **355** (2009), 277–287; arXiv:0807.3104.